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Heterocyclic compound, light-emitting device including the same, and electronic apparatus including the light-emitting device

Abstract

Provided is a heterocyclic compound represented by Formula 1, a light-emitting device including the same, and an electronic apparatus including the light-emitting device. The light-emitting device includes a first electrode, a second electrode facing the first electrode, an interlayer between the first electrode and the second electrode and including an emission layer, and the heterocyclic compound represented by Formula 1. ##STR00001##

Inventors: Um; Hyunah (Yongin-si, KR), Kim; Yunhi (Yongin-si, KR), Kwon; Soonki (Yongin-si, KR), Ahn; Heechoon (Yongin-si, KR), Park; Jihyun (Yongin-si, KR)

Applicant: Samsung Display Co., Ltd. (Yongin-si, KR); INDUSTRY-ACADEMIC COOPERATION FOUNDATION GYEONGSANG NATIONAL UNIVERSITY (Jinju-si, KR)

Family ID: 1000008782787

Assignee: Samsung Display Co., Ltd. (Yongin-si, KR); INDUSTRY-ACADEMIC COOPERATION FOUNDATION GYEONGSANG NATIONAL UNIVERSITY (Jinju-si, KR)

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Primary Examiner: Chandhok; Jenna N

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to and the benefit of Korean Patent Application No. 10-2020-0142519, filed on Oct. 29, 2020, in the Korean Intellectual Property Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Field

(2) One or more embodiments of the present disclosure relate to a heterocyclic compound, a light-emitting device including the heterocyclic compound, and an electronic apparatus including the light-emitting device.

2. Description of Related Art

(3) Among light-emitting devices, organic light-emitting devices are self-emission devices that have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed, as compared to other devices in the art.

(4) The organic light-emitting devices may include a first electrode on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode sequentially stacked on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as the holes and the electrons, recombine in the emission layer to produce excitons. These excitons transit from an excited state to a ground state to thereby generate light.

SUMMARY

(5) Provided is a light-emitting device including a compound having excellent luminance efficiency and an electronic apparatus including the light-emitting device.

(6) Additional aspects of embodiments will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

(7) According to an aspect of an embodiment, provided is a heterocyclic compound represented by Formula 1.

(8) ##STR00002##

(9) In Formula 1, EW.sub.1 may be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, EW.sub.11 may be —CN, d1 may be 0 or 1, wherein EW.sub.1 may be a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group, or d1 may be 1, T.sub.1 may be a group represented by Formula 2, and b1 may be an integer from 1 to 5, in Formula 2, X.sub.1 may be O or S, L.sub.1 is a single bond, a C.sub.4-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a1 may be an integer from 0 to 5, and Ar.sub.1 and Ar.sub.2 may each independently be a π electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a, and in Formulae 1 and 2, R.sub.1 and R.sub.2 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a

nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), or —C(=O)(Q.sub.1), c1 in Formula 1 may be an integer from 0 to 10, c2 in Formula 2 may be an integer from 0 to 7, and R.sub.10a may be: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof, or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof, and * in Formula 2 may indicate a binding site to EW.sub.1 in Formula 1.

(10) According to another aspect of an embodiment, provided is a light-emitting device including a first electrode, a second electrode facing the first electrode, an interlayer between the first electrode and the second electrode and including an emission layer, and a heterocyclic compound represented by Formula 1.

(11) According to another aspect of an embodiment, provided is an electronic apparatus including the light-emitting device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects and features of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a schematic cross-sectional view of a light-emitting device according to an

embodiment; and

(3) FIGS. 2 and 3 are each a cross-sectional view of a light-emitting apparatus according to an embodiment.

DETAILED DESCRIPTION

(4) Reference will now be made in more detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects of embodiments of the present description. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Throughout the disclosure, the expression “at least one of a, b and c” indicates only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

(5) Provided is a heterocyclic compound represented by Formula 1:

(6) ##STR00003##

(7) EW.sub.1 in Formula 1 may be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group.

(8) In an embodiment, EW.sub.1 in Formula 1 may be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a thiophene group, a furan group, an indole group, a benzoborole group, a benzophosphole group, an indene group, a benzosilole group, a benzogermole group, a benzothiophene group, a benzoselenophene group, a benzofuran group, a carbazole group, a dibenzoborole group, a dibenzophosphole group, a fluorene group, a dibenzosilole group, a dibenzogermole group, a dibenzothiophene group, a dibenzoselenophene group, a dibenzofuran group, a dibenzothiophene 5-oxide group, a 9H-fluorene-9-on group, a dibenzothiophene 5,5-dioxide group, an azaindole group, an azabenzoborole group, an azabenzophosphole group, an azaindene group, an azabenzosilole group, an azabenzogermole group, an azabenzothiophene group, an azabenzoselenophene group, an azabenzofuran group, an azacarbazole group, an azadibenzoborole group, an azadibenzophosphole group, an azafluorene group, an azadibenzosilole group, an azadibenzogermole group, an azadibenzothiophene group, an azadibenzoselenophene group, an azadibenzofuran group, an azadibenzothiophene 5-oxide group, an aza-9H-fluorene-9-on group, an azadibenzothiophene 5,5-dioxide group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

(9) EW.sub.11 in Formula 1 may be —CN.

(10) d1 in Formula 1 may be 0 or 1. d1 is the number of EW.sub.11(s).

(11) In one or more embodiments, EW.sub.1 in Formula 1 may be a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group, or d1 may be 1. For example, when EW.sub.1 is a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group, d1 may be 0 or 1, but when EW.sub.1 is not a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group, d1 may be 1.

(12) In an embodiment, EW.sub.1 in Formula 1 may be an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine group, a pyrazine group, a pyridazine group, a pyrimidine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a

phthalazine group, a naphthyridine group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, an benzoisothiazole group, a benzoxazole group, a benzoisoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, an acridine group, or a pyridopyrazine group.

(13) In one or more embodiments, EW.sub.1 in Formula 1 may be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, or a 1,2,3,4-tetrahydronaphthalene group, and d1 may be 1.

(14) In one or more embodiments, EW.sub.1 in Formula 1 may be Formula 1(a) or 1(b):

(15) ##STR00004##

(16) In Formulae 1(a) and 1(b), X.sub.11 may be N or C(R.sub.11), X.sub.12 may be N or C(R.sub.12), X.sub.13 may be N or C(R.sub.13), at least one selected from X.sub.11 to X.sub.13 may be N or C(CN), R.sub.11 to R.sub.15 are the same as described in connection with R.sub.1, and T.sub.11 and T.sub.12 are the same as described in connection with T.sub.1.

(17) In an embodiment, at least two of X.sub.11 to X.sub.13 in Formulae 1(a) and 1(b) may be N.

(18) In an embodiment, in Formulae 1(a) and 1(b), X.sub.11 may be C(R.sub.11), X.sub.12 may be C(R.sub.12), X.sub.13 may be C(R.sub.13), and at least one selected from X.sub.11 to X.sub.13 may be C(CN).

(19) b1 in Formula 1 may be an integer from 1 to 5. b1 is the number of T.sub.1(s).

(20) In an embodiment, b1 in Formula 1 may be 1 or 2.

(21) T.sub.1 in Formula 1 may be a group represented by Formula 2:

(22) ##STR00005##

(23) X.sub.1 in Formula 2 may be O or S.

(24) In one or more embodiments, X.sub.1 in Formula 2 may be O, but embodiments of the present disclosure are not limited thereto.

(25) L.sub.1 in Formula 2 may be a single bond, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

(26) In one or more embodiments, L.sub.1 in Formula 2 may be: a single bond; or a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, or a 1,2,3,4-tetrahydronaphthalene group, each unsubstituted or substituted with at least one R.sub.10a.

(27) In one or more embodiments, L.sub.1 in Formula 2 may be: a single bond; or a benzene group unsubstituted or substituted with at least one R.sub.10a, but embodiments of the present disclosure are not limited thereto.

(28) a1 in Formula 2 may be an integer from 0 to 5. a1 is the number of L.sub.1(s).

(29) In one or more embodiments, a1 in Formula 2 may be 0, 1, or 2.

(30) Ar.sub.1 and Ar.sub.2 in Formula 2 may each independently be a π electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a.

(31) In an embodiment, Ar.sub.1 and Ar.sub.2 in Formula 2 may each independently be a benzene group, a heptalene group, an indene group, a naphthalene group, an azulene group, an indacene group, an acenaphthenyl group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentacene group, a hexacene group, a pentaphene group, a rubicene group, a coronene group, an ovalene group, a pyrrole group, a furan group, a thiophene group, an isoindole group, an indole group, an indene group, a benzofuran group, a benzothiophene

group, a benzosilole group, a naphtho pyrrole group, a naphthofuran group, a naphthothiophene group, a naphthosilole group, a benzocarbazole group, a dibenzocarbazole group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, a dibenzosilole group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a triindolobenzene group, a pyrrolophenanthrene group, a furanophenanthrene group, a thienophenanthrene group, a benzonaphthofuran group, a benzonaphthothiophene group, a (indolo)phenanthrene group, a (benzofurano)phenanthrene group, or a (benzothieno)phenanthrene group, each unsubstituted or substituted with at least one R.sub.10a.

(32) In one or more embodiments, Ar.sub.1 and Ar.sub.2 in Formula 2 may each independently be a benzene group unsubstituted or substituted with at least one R.sub.10a, but embodiments of the present disclosure are not limited thereto.

(33) In Formula 1 and 2,

(34) R.sub.1 and R.sub.2 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), or —C(=O)(Q.sub.1),

(35) c1 in Formula 1 may be an integer from 0 to 10, c2 in Formula 2 may be an integer from 0 to 7, and

(36) R.sub.10a may be: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof, or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof, and * in Formula 2

may indicate a binding site to EW.sub.1 in Formula 1.

(37) In one or more embodiments, c2 in Formula 2 may be an integer from 0 to 7, where c2 indicates the number of R.sub.2(s).

(38) In an embodiment, Formula 2 may be represented by one selected from Formulae 2-1 to 2-4:

(39) ##STR00006##

(40) In Formulae 2-1 to 2-4, X.sub.1, L.sub.1, a1, Ar.sub.1, Ar.sub.2, R.sub.2, and c2 are each the same as respectively described above, and * indicates a binding site to EW.sub.1 in Formula 1. R.sub.1 and R.sub.2 in Formulae 1 and 2 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), or —C(=O)(Q.sub.1).

(41) In an embodiment, R.sub.1 and R.sub.2 in Formulae 1 and 2 may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, or a C.sub.1-C.sub.20 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or any combination thereof; a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylene group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a thienyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranlyl group, a benzothienyl group, a benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranlyl group, a dibenzothienyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, —CD.sub.3, —CD.sub.2H, —CDH.sub.2, —CF.sub.3, —CF.sub.2H, —CFH.sub.2, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.2-C.sub.20 alkenyl group, a C.sub.2-C.sub.20 alkynyl group, a C.sub.1-C.sub.20 alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a naphthyl group, a fluorenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a

thienyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothienyl group, a benzoisothiazolyl group, a benzoxazolyl group, an benzoisoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothienyl, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), —P(=O)(Q.sub.31)(Q.sub.32), or any combination thereof, or —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —N(Q.sub.1)(Q.sub.2), —B(Q.sub.1)(Q.sub.2), —C(=O)(Q.sub.1), —S(=O).sub.2(Q.sub.1), —P(=O)(Q.sub.1)(Q.sub.2), or —P(Q.sub.1)(Q.sub.2).

(42) c1 in Formula 1 may be an integer from 0 to 10. c2 in Formula 2 may be an integer from 0 to 7. c1 is the number of R.sub.1(s), and c2 is the number of R.sub.2(s).

(43) In an embodiment, the heterocyclic compound may be selected from Compounds 1 to 96 below:

(44) ##STR00007## ##STR00008## ##STR00009## ##STR00010## ##STR00011##
##STR00012## ##STR00013## ##STR00014## ##STR00015## ##STR00016## ##STR00017##
##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##

(45) The heterocyclic compound may include i) a group represented by Formula 1 and 2, and X.sub.1 in Formula 2 may be O or S. When X.sub.1 is O or S, compared to a case in which X.sub.1 is N, the heterocyclic compound may have a similar lowest unoccupied molecular orbital (LUMO) energy level and a relatively low highest occupied molecular orbital (HOMO) energy level, thereby having a large energy band gap. In the case of a group represented by Formula 1 and Formula 2, a triplet (T.sub.1) energy level is increased due to a steric hindrance effect, and thus, a difference between a singlet (S.sub.1) energy level and a triplet (T.sub.1) energy level may be increased. Also, ii) because EW.sub.1 in Formula 1 is a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group or d1 is 1, an electron-accepting capacity is enhanced and thus, electron transport characteristics are improved. Accordingly, the heterocyclic compound may be utilized as a high-efficient delayed fluorescent luminescent material, and an electronic device, for example, an organic light-emitting device, including the heterocyclic compound may have low driving voltage, excellent luminance efficiency, and a long lifespan.

(46) In one or more embodiments, the heterocyclic compound may emit blue light having a maximum emission wavelength of about 400 nm or more and about 465 nm or less, and may have a difference between a singlet (S.sub.1) energy level and a triplet (T.sub.1) energy level of 0.5 eV or less.

(47) Synthesis methods of the heterocyclic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Synthesis Examples and/or Examples provided below.

(48) At least one heterocyclic compound represented by Formula 1 may be used in a light-emitting device (for example, an organic light-emitting device). Accordingly, provided is a light-emitting device including: a first electrode; a second electrode facing the first electrode; an interlayer between the first electrode and the second electrode and including an emission layer; and a heterocyclic compound represented by Formula 1 as described herein.

(49) In an embodiment, the first electrode of the light-emitting device may be an anode, the second electrode of the light-emitting device may be a cathode, the interlayer may further include a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, the hole transport region may include a hole

injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

(50) In one or more embodiments, the heterocyclic compound may be included between the first electrode of the light-emitting device and the second electrode of the light-emitting device. In one or more embodiments, the heterocyclic compound may be included in the interlayer of the light-emitting device, for example, the emission layer or the electron transport region of the interlayer.

(51) In one or more embodiments, the emission layer in the interlayer of the light-emitting device may include a dopant and a host, and the host may include the heterocyclic compound. In other words, the heterocyclic compound may act as a host. The emission layer may emit red light, green light, blue light, and/or white light. In one or more embodiments, the emission layer may emit blue light. The blue light may have a maximum emission wavelength of, for example, about 400 nm to about 465 nm. Additional details of the emission layer and the host are the same as described elsewhere in the present specification.

(52) In one or more embodiments, the light-emitting device may include a capping layer outside the first electrode or outside the second electrode.

(53) In one or more embodiments, the light-emitting device may further include at least one selected from a first capping layer outside the first electrode and a second capping layer outside the second electrode, and at least one selected from the first capping layer and the second capping layer may include the heterocyclic compound represented by Formula 1. Further detailed description of the first capping layer and the second capping layer are the same as described elsewhere in the present specification.

(54) In an embodiment, the light-emitting device may include: a first capping layer outside the first electrode and including the heterocyclic compound represented by Formula 1; a second capping layer outside the second electrode and including the heterocyclic compound represented by Formula 1; or the first capping layer and the second capping layer.

(55) The expression “(an interlayer and/or a capping layer) includes at least one heterocyclic compound,” as used herein, may include a case in which “(an interlayer and/or a capping layer) includes identical heterocyclic compounds represented by Formula 1” and a case in which “(an interlayer and/or a capping layer) includes two or more different heterocyclic compounds represented by Formula 1.”

(56) In an embodiment, the interlayer and/or capping layer may include Compound 1 only as the heterocyclic compound. In this regard, Compound 1 may exist in the emission layer of the light-emitting device. In one or more embodiments, the interlayer may include, as the heterocyclic compound, Compound 1 and Compound 2. In this regard, Compound 1 and Compound 2 may exist in an identical layer (for example, Compound 1 and Compound 2 may all exist in an emission layer), or different layers (for example, Compound 1 may exist in an emission layer and Compound 2 may exist in an electron transport region).

(57) The term “interlayer,” as used herein, refers to a single layer and/or all layers between the first electrode and the second electrode of the light-emitting device.

(58) According to another aspect of an embodiment, provided is an electronic apparatus including the light-emitting device. The electronic apparatus may further include a thin-film transistor. In an embodiment, the electronic apparatus may further include a thin-film transistor including a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically coupled to the source electrode or the drain electrode. In one or more embodiments, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof. Further detailed description of the electronic apparatus is the same as described elsewhere in the present specification.

(59) Description of FIG. 1

(60) FIG. 1 is a schematic cross-sectional view of a light-emitting device **10** according to an embodiment. The light-emitting device **10** includes a first electrode **110**, an interlayer **130**, and a second electrode **150**.

(61) Hereinafter, a structure of the light-emitting device **10** according to an embodiment and a method of manufacturing the light-emitting device **10** will be described in connection with FIG. 1.

(62) First Electrode **110**

(63) In FIG. 1, a substrate may be additionally under the first electrode **110** or above the second electrode **150**. The substrate may be a glass substrate and/or a plastic substrate. In one or more embodiments, the substrate may be a flexible substrate. In one or more embodiments, the substrate may include plastics having excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.

(64) The first electrode **110** may be formed by, for example, depositing and/or sputtering a material for forming the first electrode **110** on the substrate. When the first electrode **110** is an anode, a high work function material that can easily inject holes may be used as a material for forming the first electrode **110**.

(65) The first electrode **110** may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode **110** is a transmissive electrode, a material for forming the first electrode **110** may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), or any combination thereof. In one or more embodiments, when the first electrode **110** is a semi-transmissive electrode or a reflective electrode, magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combination thereof may be used as a material for forming the first electrode **110**.

(66) The first electrode **110** may have a single-layered structure including (e.g., consisting of) a single layer or a multi-layered structure including a plurality of layers. In an embodiment, the first electrode **110** may have a three-layered structure of ITO/Ag/ITO.

(67) Interlayer **130**

(68) The interlayer **130** is on the first electrode **110**. The interlayer **130** includes an emission layer.

(69) The interlayer **130** may further include a hole transport region between the first electrode **110** and the emission layer and an electron transport region between the emission layer and the second electrode **150**.

(70) The interlayer **130** may further include metal-containing compounds such as organometallic compounds, inorganic materials such as quantum dots, and/or the like, in addition to various suitable organic materials.

(71) In one or more embodiments, the interlayer **130** may include, i) two or more emitting units sequentially stacked between the first electrode **110** and the second electrode **150** and ii) a charge generation layer between the two emitting units. When the interlayer **130** includes the emitting units and the charge generation layer, the light-emitting device **10** may be a tandem light-emitting device.

(72) Hole Transport Region in Interlayer **130**

(73) The hole transport region may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

(74) The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.

(75) For example, the hole transport region may have a multi-layered structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport

layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/electron blocking layer structure, wherein, in each structure, layers are stacked sequentially from the first electrode **110**.

(76) The hole transport region may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:

(77) ##STR00030##

(78) In Formulae 201 and 202, L.sub.201 to L.sub.204 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, L.sub.205 may be *—O—*, *—S—*, *—N(Q.sub.201)—*, a C.sub.1-C.sub.20 alkylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.20 alkenylene group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xa1 to xa4 may each independently be an integer from 0 to 5, xa5 may be an integer from 1 to 10, R.sub.201 to R.sub.204 and Q.sub.201 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, R.sub.201 and R.sub.202 may optionally be linked to each other, via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 polycyclic group unsubstituted or substituted with at least one R.sub.10a (for example, a carbazole group or the like) (for example, refer to the following Compound HT16), R.sub.203 and R.sub.204 may optionally be linked to each other, via a single bond, a C.sub.1-C.sub.5 alkylene group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.2-C.sub.5 alkenylene group unsubstituted or substituted with at least one R.sub.10a, to form a C.sub.8-C.sub.60 poly cyclic group unsubstituted or substituted with at least one R.sub.10a, and na1 may be an integer from 1 to 4.

(79) In an embodiment, Formulae 201 and 202 may each include at least one selected from groups represented by Formulae CY201 to CY217:

(80) ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036## ##STR00037##

(81) Regarding Formulae CY201 to CY217, R.sub.10b and R.sub.10c are the same as described in connection with R.sub.10a, ring CY.sub.201 to ring CY.sub.204 may each independently be a C.sub.3-C.sub.20 carbocyclic group or a C.sub.1-C.sub.20 heterocyclic group, and at least one hydrogen in Formula CY201 to CY217 may be unsubstituted or substituted with at least one R.sub.10a described herein.

(82) In an embodiment, ring CY.sub.201 to ring CY.sub.204 in Formulae CY201 to CY217 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.

(83) In an embodiment, Formulae 201 and 202 may each include at least one selected from groups represented by Formulae CY201 to CY203.

(84) In one or more embodiments, Formula 201 may include at least one selected from groups represented by Formulae CY201 to CY203 and at least one selected from groups represented by Formulae CY204 to CY217.

(85) In one or more embodiments, in Formula 201, xa1 may be 1, R.sub.201 may be a group represented by one selected from Formulae CY201 to CY203, xa2 may be 0, and R.sub.202 may be a group represented by one selected from Formulae CY204 to CY207.

(86) In one or more embodiments, each of Formulae 201 and 202 may not include a group represented by one selected from Formulae CY201 to CY203.

(87) In one or more embodiments, each of Formulae 201 and 202 may not include a group

represented by one selected from Formulae CY201 to CY203 and may include at least one selected from groups represented by Formulae CY204 to CY217.

(88) In an embodiment, each of Formulae 201 and 202 may not include a group represented by one selected from Formulae CY201 to CY217.

(89) In an embodiment, the hole transport region may include one selected from Compounds HT1 to HT46, m-MTDATA, TDATA, 2-TNATA, NPB(NPD), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4',4''-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), or any combination thereof:

(90) ##STR00038## ##STR00039## ##STR00040## ##STR00041## ##STR00042##

##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047## ##STR00048##

(91) A thickness of the hole transport region may be in a range of about 50 Å to about 10,000 Å, for example, about 100 Å to about 4,000 Å. When the hole transport region includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, for example, about 100 Å to about 1,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å, for example, about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, suitable or satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

(92) The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance according to the wavelength of light emitted by an emission layer, and the electron blocking layer may prevent or reduce leakage of electrons from an emission layer to a hole transport region. Materials that may be included in the hole transport region may be included in the emission auxiliary layer and the electron blocking layer.

(93) p-Dopant

(94) The hole transport region may further include, in addition to these materials, a charge-generation material for improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region (for example, in the form of a single layer including (e.g., consisting of) a charge-generation material).

(95) The charge-generation material may be, for example, a p-dopant.

(96) In an embodiment, the lowest unoccupied molecular orbital (LUMO) energy level of the p-dopant may be -3.5 eV or less.

(97) In an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound containing element EL1 and element EL2, or any combination thereof.

(98) Examples of the quinone derivative may include TCNQ, F4-TCNQ, etc.

(99) Examples of the cyano group-containing compound may include HAT-CN and a compound represented by Formula 221 below.

(100) ##STR00049##

(101) In Formula 221, R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, and at least one selected from R.sub.221 to R.sub.223 may each independently be a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each substituted with: a cyano group; —F; —Cl; —Br; —I; a C.sub.1-C.sub.20 alkyl group substituted with a cyano group, —F, —Cl, —Br, —I, or any combination thereof; or any combination thereof.

(102) Regarding the compound containing element EL1 and element EL2, element EL1 may be metal, metalloid, or a combination thereof, and element EL2 may be a non-metal, metalloid, or a combination thereof.

(103) Examples of the metal may include: an alkali metal (for example, lithium (Li), sodium (Na),

potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (for example, beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal (for example, titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium (Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag), gold (Au), etc.); a post-transition metal (for example, zinc (Zn), indium (In), tin (Sn), etc.); and a lanthanide metal (for example, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.).

(104) Examples of the metalloid may include silicon (Si), antimony (Sb), and tellurium (Te).

(105) Examples of the non-metal may include oxygen (O) and halogen (for example, F, Cl, Br, I, etc.).

(106) In an embodiment, examples of the compound containing element EL1 and element EL2 may include metal oxide, metal halide (for example, metal fluoride, metal chloride, metal bromide, and/or metal iodide), metalloid halide (for example, metalloid fluoride, metalloid chloride, metalloid bromide, and/or metalloid iodide), metal telluride, or any combination thereof.

(107) Examples of the metal oxide may include tungsten oxide (for example, WO, W.sub.2O.sub.3, WO.sub.2, WO.sub.3, and/or W.sub.2O.sub.5), vanadium oxide (for example, VO, V.sub.2O.sub.3, VO.sub.2, and/or V.sub.2O.sub.5), molybdenum oxide (MoO, Mo.sub.2O.sub.3, MoO.sub.2, MoO.sub.3, and/or Mo.sub.2O.sub.5), and rhenium oxide (for example, ReO.sub.3).

(108) Examples of the metal halide may include alkali metal halide, alkaline earth metal halide, transition metal halide, post-transition metal halide, and lanthanide metal halide.

(109) Examples of the alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, LiI, NaI, KI, RbI, and CsI.

(110) Examples of the alkaline earth metal halide may include BeF.sub.2, MgF.sub.2, CaF.sub.2, SrF.sub.2, BaF.sub.2, BeCl.sub.2, MgCl.sub.2, CaCl.sub.2, SrCl.sub.2, BaCl.sub.2, BeBr.sub.2, MgBr.sub.2, CaBr.sub.2, SrBr.sub.2, BaBr.sub.2, BeI.sub.2, MgI.sub.2, CaI.sub.2, SrI.sub.2, and BaI.sub.2.

(111) Examples of the transition metal halide may include titanium halide (for example, TiF.sub.4, TiCl.sub.4, TiBr.sub.4, and/or TiI.sub.4), zirconium halide (for example, ZrF.sub.4, ZrCl.sub.4, ZrBr.sub.4, and/or ZrI.sub.4), hafnium halide (for example, HfF.sub.4, HfCl.sub.4, HfBr.sub.4, and/or HfI.sub.4), vanadium halide (for example, VF.sub.3, VCl.sub.3, VBr.sub.3, and/or VI.sub.3), niobium halide (for example, NbF.sub.3, NbCl.sub.3, NbBr.sub.3, and/or NbI.sub.3), tantalum halide (for example, TaF.sub.3, TaCl.sub.3, TaBr.sub.3, and/or TaI.sub.3), chromium halide (for example, CrF.sub.3, CrCl.sub.3, CrBr.sub.3, and/or CrI.sub.3), molybdenum halide (for example, MoF.sub.3, MoCl.sub.3, MoBr.sub.3, and/or MoI.sub.3), tungsten halide (for example, WF.sub.3, WCl.sub.3, WBr.sub.3, and/or WI.sub.3), manganese halide (for example, MnF.sub.2, MnCl.sub.2, MnBr.sub.2, and/or MnI.sub.2), technetium halide (for example, TcF.sub.2, TcCl.sub.2, TcBr.sub.2, and/or TcI.sub.2), rhenium halide (for example, ReF.sub.2, ReCl.sub.2, ReBr.sub.2, and/or ReI.sub.2), iron halide (for example, FeF.sub.2, FeCl.sub.2, FeBr.sub.2, and/or FeI.sub.2), ruthenium halide (for example, RuF.sub.2, RuCl.sub.2, RuBr.sub.2, and/or RuI.sub.2), osmium halide (for example, OsF.sub.2, OsCl.sub.2, OsBr.sub.2, and/or OsI.sub.2), cobalt halide (for example, CoF.sub.2, CoCl.sub.2, CoBr.sub.2, and/or CoI.sub.2), rhodium halide (for example, RhF.sub.2, RhCl.sub.2, RhBr.sub.2, and/or RhI.sub.2), iridium halide (for example, IrF.sub.2, IrCl.sub.2, IrBr.sub.2, and/or IrI.sub.2), nickel halide (for example, NiF.sub.2, NiCl.sub.2, NiBr.sub.2, and/or NiI.sub.2), palladium halide (for example, PdF.sub.2, PdCl.sub.2, PdBr.sub.2, and/or PdI.sub.2), platinum halide (for example, PtF.sub.2, PtCl.sub.2, PtBr.sub.2, and/or PtI.sub.2), copper halide (for example, CuF, CuCl, CuBr, and/or CuI), silver halide (for example, AgF, AgCl, AgBr, and/or AgI), and gold halide (for example, AuF, AuCl, AuBr, and/or AuI).

- (112) Examples of the post-transition metal halide may include zinc halide (for example, ZnF.sub.2, ZnCl.sub.2, ZnBr.sub.2, and/or ZnI.sub.2), indium halide (for example, InI.sub.3), and tin halide (for example, SnI.sub.2).
- (113) Examples of the lanthanide metal halide may include YbF, YbF.sub.2, YbF.sub.3, SmF.sub.3, YbCl, YbCl.sub.2, YbCl.sub.3, SmCl.sub.3, YbBr, YbBr.sub.2, YbBr.sub.3, SmBr.sub.3, YbI, YbI.sub.2, YbI.sub.3, and SmI.sub.3.
- (114) Examples of the metalloid halide may include antimony halide (for example, SbCl.sub.5).
- (115) Examples of the metal telluride may include alkali metal telluride (for example, Li.sub.2Te, Na.sub.2Te, K.sub.2Te, Rb.sub.2Te, and/or Cs.sub.2Te), alkaline earth metal telluride (for example, BeTe, MgTe, CaTe, SrTe, and/or BaTe), transition metal telluride (for example, TiTe.sub.2, ZrTe.sub.2, HfTe.sub.2, V.sub.2Te.sub.3, Nb.sub.2Te.sub.3, Ta.sub.2Te.sub.3, Cr.sub.2Te.sub.3, Mo.sub.2Te.sub.3, W.sub.2Te.sub.3, MnTe, TcTe, ReTe, FeTe, RuTe, OsTe, CoTe, RhTe, IrTe, NiTe, PdTe, PtTe, Cu.sub.2Te, CuTe, Ag.sub.2Te, AgTe, and/or Au.sub.2Te), post-transition metal telluride (for example, ZnTe), and lanthanide metal telluride (for example, LaTe, CeTe, PrTe, NdTe, PmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, and/or LuTe).
- (116) Emission Layer in Interlayer **130**
- (117) When the light-emitting device **10** is a full-color light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer, according to a sub-pixel. In one or more embodiments, the emission layer may have a stacked structure of two or more layers of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers contact each other or are separated from each other to emit white light. In one or more embodiments, the emission layer may include two or more materials of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.
- (118) The emission layer may include a host and a dopant. The dopant may include a phosphorescent dopant, a fluorescent dopant, or any combination thereof.
- (119) The amount of the dopant in the emission layer may be from about 0.01 to about 15 parts by weight based on 100 parts by weight of the host.
- (120) In one or more embodiments, the emission layer may include a quantum dot.
- (121) In one or more embodiments, the emission layer may include a delayed fluorescent material. The delayed fluorescent material may act as a host or a dopant in the emission layer.
- (122) A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å, for example, about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage.
- (123) Host
- (124) The host may include the compound represented by Formula 1, a compound represented by Formula 301, or any combination thereof:

$$[\text{Ar.sub.301}]_{\text{sub.xb11}} - [(\text{L.sub.301})_{\text{sub.xb1}} - \text{R.sub.301}]_{\text{sub.xb21}} \quad \text{Formula 301}$$
- (125) In Formula 301, Ar.sub.301 and L.sub.301 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xb11 may be 1, 2, or 3, xb1 may be an integer from 0 to 5, R.sub.301 may be hydrogen, deuterium, —F, —Cl, —Br, —I, hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.301)(Q.sub.302)(Q.sub.303), —N(Q.sub.301)

(Q.sub.302), —B(Q.sub.301)(Q.sub.302), —C(=O)(Q.sub.301), —S(=O).sub.2(Q.sub.301), or —P(=O)(Q.sub.301)(Q.sub.302), xb21 may be an integer from 1 to 5, and Q.sub.301 to Q.sub.303 are the same as described in connection with Q.sub.1.

(126) In one or more embodiments, when xb11 in Formula 301 is 2 or more, two or more of Ar.sub.301(s) may be linked to each other via a single bond.

(127) In an embodiment, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:

(128) ##STR00050##

(129) In Formulae 301-1 and 301-2, ring A.sub.301 to ring A.sub.304 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, X.sub.301 may be O, S, N-[(L.sub.304).sub.xb4-R.sub.304], C(R.sub.304)(R.sub.305), or Si(R.sub.304)(R.sub.305), xb22 and xb23 may each independently be 0, 1, or 2, L.sub.301, xb1, and R.sub.301 may each be the same as respectively described elsewhere in the present specification, L.sub.302 to L.sub.304 may each independently be the same as respectively described in connection with L.sub.301, xb2 to xb4 may each independently be the same as respectively described in connection with xb1, and R.sub.302 to R.sub.305 and R.sub.311 to R.sub.314 may each be the same as respectively described in connection with R.sub.301.

(130) In one or more embodiments, the host may include an alkaline earth-metal complex, a post-transition metal complex, or any combination thereof. In an embodiment, the host may include a Be complex (for example, Compound H55), a Mg complex, a Zn complex, or any combination thereof.

(131) In an embodiment, the host may include one selected from Compounds H1 to H124, 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), or any combination thereof, but embodiments of the present disclosure are not limited thereto:

(132) ##STR00051## ##STR00052## ##STR00053## ##STR00054## ##STR00055##
##STR00056## ##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061##
##STR00062## ##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067##
##STR00068## ##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073##
##STR00074## ##STR00075## ##STR00076## ##STR00077## ##STR00078## ##STR00079##
##STR00080##

Phosphorescent Dopant

(133) The phosphorescent dopant may include at least one transition metal as a central metal.

(134) The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.

(135) The phosphorescent dopant may be electrically neutral.

(136) In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

(137) ##STR00081##

(138) In Formulae 401 and 402, M may be transition metal (for example, iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)), L.sub.401 may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is 2 or more, two or more of L.sub.401(s) may be identical to or different from each other, L.sub.402 may be an organic ligand, and xc2 may be 0, 1, 2, 3, or 4, and when xc2 is 2 or more, two or more of L.sub.402(s) may be identical to or different from each other, X.sub.401 and X.sub.402 may each independently be nitrogen or carbon, ring A.sub.401 and ring A.sub.402 may each independently be a C.sub.3-

C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, T.sub.401 may be a single bond, *—O—*, *—S—*, *—C(=O)—*, *—N(Q.sub.411)—*, *—C(Q.sub.411)(Q.sub.412)—*, *—C(Q.sub.411)=C(Q.sub.412)—*, *—C(Q.sub.411)=*, or *=C=*, X.sub.403 and X.sub.404 may each independently be a chemical bond (for example, a covalent bond or a coordination bond), O, S, N(Q.sub.413), B(Q.sub.413), P(Q.sub.413), C(Q.sub.413)(Q.sub.414), or Si(Q.sub.413)(Q.sub.414), Q.sub.411 to Q.sub.414 may each be the same as respectively described in connection with Q.sub.1, R.sub.401 and R.sub.402 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.20 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.401)(Q.sub.402)(Q.sub.403), —N(Q.sub.401)(Q.sub.402), —B(Q.sub.401)(Q.sub.402), —C(=O)(Q.sub.401), —S(=O).sub.2(Q.sub.401), or —P(=O)(Q.sub.401)(Q.sub.402), Q.sub.401 to Q.sub.403 may each be the same as respectively described in connection with Q.sub.1, xc11 and xc12 may each independently be an integer from 0 to 10, and * and *' in Formula 402 each indicate a binding site to M in Formula 401.

(139) In an embodiment, in Formula 402, i) X.sub.401 may be nitrogen, and X.sub.402 may be carbon, or ii) each of X.sub.401 and X.sub.402 may be nitrogen.

(140) In an embodiment, when xc1 in Formula 401 is 2 or more, two ring A.sub.401 in two or more of L.sub.401(s) may be optionally linked to each other via T.sub.402, which is a linking group, and two ring A.sub.402 may optionally be linked to each other via T.sub.403, which is a linking group (see Compounds PD1 to PD4 and PD7). T.sub.402 and T.sub.403 may each be the same as respectively described in connection with T.sub.401.

(141) L.sub.402 in Formula 401 may be an organic ligand. In an embodiment, L.sub.402 may include a halogen group, a diketone group (for example, an acetylacetonate group), a carboxylic acid group (for example, a picolinate group), —C(=O), an isonitrile group, a —CN group, a phosphorus group (for example, a phosphine group, a phosphite group, etc.), or any combination thereof.

(142) The phosphorescent dopant may include, for example, one selected from compounds PD1 to PD25 or any combination thereof:

(143) ##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086## ##STR00087##

Fluorescent Dopant

(144) The fluorescent dopant may include an amine group-containing compound, a styryl group-containing compound, or any combination thereof.

(145) In an embodiment, the fluorescent dopant may include a compound represented by Formula 501:

(146) ##STR00088##

(147) In Formula 501, Ar.sub.501, L.sub.501 to L.sub.503, R.sub.501 and R.sub.502 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xd1 to xd3 may each independently be 0, 1, 2, or 3, and xd4 may be 1, 2, 3, 4, 5, or 6.

(148) In an embodiment, Ar.sub.501 in Formula 501 may include a condensed cyclic group (for example, an anthracene group, a chrysene group, or a pyrene group) in which three or more monocyclic groups are condensed together (e.g., combined together).

(149) In one or more embodiments, xd4 in Formula 501 may be 2.

(150) In an embodiment, the fluorescent dopant may include: one selected from Compounds FD1 to FD36; DPVBi; DPAVBi; or any combination thereof:

(151) ##STR00089## ##STR00090## ##STR00091## ##STR00092## ##STR00093##
##STR00094##

Delayed Fluorescent Material

(152) The emission layer may include a delayed fluorescent material.

(153) The delayed fluorescence material used herein may be selected from the heterocyclic compound represented by Formula 1 or any suitable compound that may emit delayed fluorescence by a delayed fluorescence emission mechanism.

(154) The delayed fluorescent material included in the emission layer may act as a host or a dopant depending on the type or kind of other materials included in the emission layer.

(155) In an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescent material and a singlet energy level (eV) of the delayed fluorescent material may be 0 eV or more and 0.5 eV or less. When the difference between the triplet energy level (eV) of the delayed fluorescent material and the singlet energy level (eV) of the delayed fluorescent material satisfies the above-described range, up-conversion from a triplet state to a singlet state of the delayed fluorescent materials may effectively occur, and thus, the luminescence efficiency of the light-emitting device **10** may be improved.

(156) In an embodiment, the delayed fluorescent material may include i) a material that includes at least one electron donor (for example, a π electron-rich C.sub.3-C.sub.60 cyclic group, such as a carbazole group) and at least one electron acceptor (for example, a sulfoxide group, a cyano group, or a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group), ii) a material including a C.sub.8-C.sub.60 polycyclic group in which two or more cyclic groups share boron (B) and are condensed with (e.g., combined together with) each other.

(157) The delayed fluorescent material may include at least one selected from Compounds DF1 to DF9:

(158) ##STR00095## ##STR00096## ##STR00097##

Quantum Dot

(159) The emission layer may include a quantum dot.

(160) The quantum dot used herein refers to a crystal of a semiconductor compound, and may include any suitable material that is capable of emitting light of various suitable emission wavelengths depending on a size of the crystal.

(161) A diameter of the quantum dot may be, for example, in a range of about 1 nm to about 10 nm.

(162) The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or a process that is similar to these processes.

(163) The wet chemical process refers to a method in which a solvent and a precursor material are mixed, and then, a quantum dot particle crystal is grown. When the crystal grows, the organic solvent naturally acts as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles can be controlled through a process which is more easily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE), and which requires low costs.

(164) The quantum dot may include Groups II-VI semiconductor compound, Groups III-V semiconductor compound, Groups III-VI semiconductor compound, Groups I-III-VI semiconductor compound, Groups IV-VI semiconductor compound, a Group IV element or compound, or any combination thereof.

(165) Examples of the Groups II-VI semiconductor compound may include: a binary compound, such as CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, and/or MgS; a ternary compound, such as CdSeS, CdSeTe, CdSTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, and/or MgZnS; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe,

CdHgSTe, HgZnSeS, HgZnSeTe, and/or HgZnSTe; or any combination thereof.

(166) Examples of the Groups III-V semiconductor compound may include: a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AlP, AlAs, AlSb, InN, InP, InAs, and/or InSb; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAs, GaPSb, AlNP, AlNAs, AlNSb, AlPAs, AlPSb, InGaP, InNP, InAlP, InNAs, InNSb, InPAs, and/or InPSb; a quaternary compound, such as GaAlNAs, GaAlNSb, GaAlPAs, GaAlPSb, GaInNP, GaInNAs, GaInNSb, GaInPAs, GaInPSb, InAlNP, InAlNAs, InAlNSb, InAlPAs, InAlPSb, and/or GaAlNP; or any combination thereof. In one or more embodiments, the Groups III-V semiconductor compound may further include a Group II element. Examples of the Groups III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, and/or InAlZnP.

(167) Examples of the Groups III-VI semiconductor compound may include: a binary compound, such as GaS, GaSe, Ga.sub.2Se.sub.3, GaTe, InS, In.sub.2S.sub.3, InSe, In.sub.2Se.sub.3, and/or InTe; a ternary compound, such as InGaS.sub.3, and/or InGaSe.sub.3; or any combination thereof.

(168) Examples of the Group I-III-VI semiconductor compound may include: a ternary compound, such as AgInS, AgInS.sub.2, CuInS, CuInS.sub.2, CuGaO.sub.2, AgGaO.sub.2, and/or AgAlO.sub.2.

(169) Examples of the Group IV-VI semiconductor compound may include: a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, and/or PbTe; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, and/or SnPbTe; a quaternary compound, such as SnPbSSe, SnPbSeTe, and/or SnPbSTe; or any combination thereof.

(170) In an embodiment, the Group IV element or compound may include: a single element, such as Si and/or Ge; a binary compound, such as SiC and/or SiGe; or any combination thereof.

(171) Each element included in the multi-element compound such as the binary compound, ternary compound, and quaternary compound may be present in a particle at a uniform concentration or a non-uniform concentration.

(172) In an embodiment, the quantum dot may have a single structure having a uniform (e.g., substantially uniform) concentration of each element included in the corresponding quantum dot or a dual structure of a core-shell. In an embodiment, the material included in the core may be different from the material included in the shell.

(173) The shell of the quantum dot may function as a protective layer for maintaining semiconductor characteristics by preventing or reducing chemical degeneration of the core and/or may function as a charging layer for imparting electrophoretic characteristics to the quantum dot. The shell may be a single layer or a multilayer. An interface between the core and the shell may have a concentration gradient in which the concentration of elements existing in the shell decreases along a direction toward the center.

(174) Examples of the shell of the quantum dot may include an oxide of metal, metalloid, and/or non-metal, a semiconductor compound, or any combination thereof. Examples of the oxide of metal, metalloid, and/or non-metal may include a binary compound, such as SiO.sub.2, Al.sub.2O.sub.3, TiO.sub.2, ZnO, MnO, Mn.sub.2O.sub.3, Mn.sub.3O.sub.4, CuO, FeO, Fe.sub.2O.sub.3, Fe.sub.3O.sub.4, CoO, Co.sub.3O.sub.4, and/or NiO; a ternary compound, such as MgAl.sub.2O.sub.4, CoFe.sub.2O.sub.4, NiFe.sub.2O.sub.4, and/or CoMn.sub.2O.sub.4; and any combination thereof. Examples of the semiconductor compound may include, as described herein, Groups II-VI semiconductor compound, Groups III-V semiconductor compound, Groups III-VI semiconductor compound, Groups I-III-VI semiconductor compound, Groups IV-VI semiconductor compound, or any combination thereof. In an embodiment, the semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AlP, AlSb, or any combination thereof.

(175) A full width at half maximum (FWHM) of an emission wavelength spectrum of the quantum dot may be about 45 nm or less, for example, about 40 nm or less, for example, about 30 nm or less. When the FWHM of the emission wavelength spectrum of the quantum dot is within this

range, color purity or color reproduction may be improved. In addition, light emitted through such quantum dots is irradiated in omnidirection (e.g., substantially every direction). Accordingly, a wide viewing angle may be increased.

(176) In addition, the quantum dot may be, for example, a spherical nanoparticle, a pyramidal nanoparticle, a multi-arm nanoparticle, a cubic nanoparticle, a nanotube particle, a nanowire particle, a nanofiber particle, and/or a nanoplate particle.

(177) Because the energy band gap can be adjusted by controlling the size of the quantum dot, light having various suitable wavelength bands can be obtained from the quantum dot emission layer. Therefore, by using quantum dots of different sizes, a light-emitting device that emits light of various suitable wavelengths may be implemented. In more detail, the size of the quantum dot may be selected to emit red, green and/or blue light. In addition, the size of the quantum dot may be configured to emit white light by combining light of various suitable colors.

(178) Electron Transport Region in Interlayer **130**

(179) The electron transport region may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

(180) The electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

(181) In an embodiment, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein, for each structure, constituting layers are sequentially stacked from an emission layer.

(182) The electron transport region (for example, the buffer layer, the hole blocking layer, the electron control layer, or the electron transport layer in the electron transport region) may include a metal-free compound including at least one π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group.

(183) In an embodiment, the electron transport region may include the heterocyclic compound represented by Formula 1, a compound represented by Formula 601, or any combination thereof.

[Ar.sub.601].sub.xe11-[(L.sub.601).sub.xe1-R.sub.601].sub.xe21 Formula 601

(184) In Formula 601, Ar.sub.601 and L.sub.601 may each independently be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, xe11 may be 1, 2, or 3, xe1 may be 0, 1, 2, 3, 4, or 5, R.sub.601 may be a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.601)(Q.sub.602)(Q.sub.603), —C(=O)(Q.sub.601), —S(=O).sub.2(Q.sub.601), or —P(=O)(Q.sub.601)(Q.sub.602), Q.sub.601 to Q.sub.603 are the same as described in connection with Q.sub.1, xe21 may be 1, 2, 3, 4, or 5, and at least one selected from Ar.sub.601, L.sub.601, and R.sub.601 may each independently be a π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a.

(185) In one or more embodiments, when xe11 in Formula 601 is 2 or more, two or more of Ar.sub.601(s) may be linked to each other via a single bond.

(186) In an embodiment, Ar.sub.601 in Formula 601 may be a substituted or unsubstituted anthracene group.

(187) In an embodiment, the electron transport region may include a compound represented by Formula 601-1:

(188) ##STR00098##

(189) In Formula 601-1, X.sub.614 may be N or C(R.sub.614), X.sub.615 may be N or

C(R.sub.615), X.sub.616 may be N or C(R.sub.616), and at least one selected from X.sub.614 to X.sub.616 may be N, L.sub.611 to L.sub.613 may be understood by referring to the description presented in connection with L.sub.601, xe611 to xe613 may be understood by referring to the description presented in connection with xe1, R.sub.611 to R.sub.613 may be understood by referring to the description presented in connection with R.sub.601, and R.sub.614 to R.sub.616 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.20 alkyl group, a C.sub.1-C.sub.20 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

(190) In an embodiment, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

(191) The electron transport region may include one selected from Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq.sub.3, BAlq, TAZ, NTAZ, or any combination thereof:

(192) ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103##
##STR00104## ##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109##
##STR00110## ##STR00111## ##STR00112##

(193) A thickness of the electron transport region may be from about 100 Å to about 5,000 Å, for example, about 160 Å to about 4,000 Å. When the electron transport region includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be from about 20 Å to about 1,000 Å, for example, about 30 Å to about 300 Å, and a thickness of the electron transport layer may be from about 100 Å to about 1,000 Å, for example, about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, the electron transport layer, and/or the electron transport region are within these ranges, suitable or satisfactory electron transport characteristics may be obtained without a substantial increase in driving voltage.

(194) The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

(195) The metal-containing material may include an alkali metal complex, an alkaline earth-metal complex, or any combination thereof. A metal ion of the alkali metal complex may be a Li ion, a Na ion, a K ion, a Rb ion, or a Cs ion, and a metal ion of the alkaline earth-metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may each independently be a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a hydroxy phenyloxadiazole, a hydroxy phenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

(196) In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (Liq) or ET-D2:

(197) ##STR00113##

(198) The electron transport region may include an electron injection layer that facilitates the injection of electrons from the second electrode **150**. The electron injection layer may be in direct contact (e.g., physical contact) with the second electrode **150**.

(199) The electron injection layer may have: i) a single-layered structure including (e.g., consisting of) a single layer including (e.g., consisting of) a single material, ii) a single-layered structure including (e.g., consisting of) a single layer including a plurality of different materials, or iii) a multi-layered structure including a plurality of layers including different materials.

(200) The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

(201) The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.

(202) The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may be oxides and/or halides (for example, fluorides, chlorides, bromides, and/or iodides) of the alkali metal, the alkaline earth metal, and the rare earth metal, telluride, or any combination thereof.

(203) The alkali metal-containing compound may be alkali metal oxides, such as $\text{Li}_{2\text{O}}$, $\text{Cs}_{2\text{O}}$, and/or $\text{K}_{2\text{O}}$, alkali metal halides, such as LiF, NaF, CsF, KF, LiI, NaI, CsI, and/or KI, or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxide, such as BaO, SrO, CaO, $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ (x is a real number that satisfies the condition of $0 < x < 1$), and/or $\text{Ba}_x\text{Ca}_{1-x}\text{O}$ (x is a real number that satisfies the condition of $0 < x < 1$). The rare earth metal-containing compound may include YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , TbF_3 , YbI_3 , ScI_3 , TbI_3 , or any combination thereof. In an embodiment, the rare earth metal-containing compound may include lanthanide metal telluride. Examples of the lanthanide metal telluride may include LaTe, CeTe, PrTe, NdTe, PmTe, SmTe, EuTe, GdTe, TbTe, DyTe, HoTe, ErTe, TmTe, YbTe, LuTe, $\text{La}_{2/3}\text{Te}_3$, $\text{Ce}_{2/3}\text{Te}_3$, $\text{Pr}_{2/3}\text{Te}_3$, $\text{Nd}_{2/3}\text{Te}_3$, $\text{Pm}_{2/3}\text{Te}_3$, $\text{Sm}_{2/3}\text{Te}_3$, $\text{Eu}_{2/3}\text{Te}_3$, $\text{Gd}_{2/3}\text{Te}_3$, $\text{Tb}_{2/3}\text{Te}_3$, $\text{Dy}_{2/3}\text{Te}_3$, $\text{Ho}_{2/3}\text{Te}_3$, $\text{Er}_{2/3}\text{Te}_3$, $\text{Tm}_{2/3}\text{Te}_3$, $\text{Yb}_{2/3}\text{Te}_3$, and $\text{Lu}_{2/3}\text{Te}_3$.

(204) The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include i) one selected from metal ions of the alkali metal, the alkaline earth metal, and the rare earth metal and ii) as a ligand linked to the metal ion, for example, a hydroxy quinoline, a hydroxy isoquinoline, a hydroxy benzoquinoline, a hydroxy acridine, a hydroxy phenanthridine, a hydroxy phenyloxazole, a hydroxy phenylthiazole, a hydroxy phenyloxadiazole, a hydroxy phenylthiadiazole, a hydroxy phenylpyridine, a hydroxy phenylbenzimidazole, a hydroxy phenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

(205) The electron injection layer may include (e.g., consist of) an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof, or may further include an organic material (for example, a compound represented by Formula 601).

(206) In an embodiment, the electron injection layer may include (e.g., consist of) i) an alkali metal-containing compound (for example, an alkali metal halide), or ii) a) an alkali metal-containing compound (for example, an alkali metal halide); and b) alkali metal, alkaline earth metal, rare earth metal, or any combination thereof. In an embodiment, the electron injection layer may be a KI:Yb co-deposited layer and/or a RbI:Yb co-deposited layer.

(207) When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

(208) A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å, for example, about 3 Å to about 90 Å. When the thickness of the electron injection layer is within the

range described above, the electron injection layer may have suitable or satisfactory electron injection characteristics without a substantial increase in driving voltage.

(209) Second Electrode **150**

(210) The second electrode **150** may be on the interlayer **130** having such a structure. The second electrode **150** may be a cathode, which is an electron injection electrode, and as a material for forming the second electrode **150**, a metal, an alloy, an electrically conductive compound, or any combination thereof, each having a low work function, may be utilized.

(211) The second electrode **150** may include lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ytterbium (Yb), silver-ytterbium (Ag—Yb), ITO, IZO, or any combination thereof. The second electrode **150** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

(212) The second electrode **150** may have a single-layered structure or a multi-layered structure including two or more layers.

(213) Capping Layer

(214) A first capping layer may be outside the first electrode **110**, and/or a second capping layer may be outside the second electrode **150**. In more detail, the light-emitting device **10** may have a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, and the second electrode **150** are sequentially stacked in this stated order, a structure in which the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are sequentially stacked in this stated order, or a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are sequentially stacked in this stated order.

(215) Light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the first electrode **110**, which is a semi-transmissive electrode or a transmissive electrode, and the first capping layer, and light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the second electrode **150**, which is a semi-transmissive electrode or a transmissive electrode, and the second capping layer.

(216) The first capping layer and the second capping layer may increase external luminescence efficiency according to the principle of constructive interference. Accordingly, the optical extraction efficiency of the light-emitting device **10** is increased, so that the luminescence efficiency of the light-emitting device **10** may be improved.

(217) Each of the first capping layer and the second capping layer may include a material having a refractive index of 1.6 or more (at a wavelength of 589 nm).

(218) The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic material including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.

(219) At least one selected from the first capping layer and the second capping layer may each independently include a carbocyclic compound, a heterocyclic compound, an amine group-containing compound, a porphyrine derivative, a phthalocyanine derivative, a naphthalocyanine derivative, an alkali metal complex, an alkaline earth-metal complex, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing compound may be optionally substituted with a substituent containing O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof. In an embodiment, at least one selected from the first capping layer and the second capping layer may each independently include an amine group-containing compound.

(220) In an embodiment, at least one selected from the first capping layer and second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.

(221) In one or more embodiments, at least one selected from the first capping layer and the second capping layer may each independently include one selected from Compounds HT28 to HT33, one selected from Compounds CP1 to CP6, β -NPB, or any combination thereof:

(222) ##STR00114## ##STR00115##

Electronic Apparatus

(223) The light-emitting device may be included in various suitable electronic apparatuses. In an embodiment, the electronic apparatus including the light-emitting device may be a light-emitting apparatus, an authentication apparatus, and/or the like.

(224) The electronic apparatus (for example, light-emitting apparatus) may further include, in addition to the light-emitting device, i) a color filter, ii) a color conversion layer, or iii) a color filter and a color conversion layer. The color filter and/or the color conversion layer may be in at least one traveling direction of light emitted from the light-emitting device. In an embodiment, light emitted from the light-emitting device may be blue light and/or white light. The light-emitting device may be the same as described above. In an embodiment, the color conversion layer may include a quantum dot. The quantum dot may be, for example, a quantum dot as described elsewhere herein.

(225) The electronic apparatus may include a first substrate. The first substrate includes a plurality of subpixel areas, the color filter includes a plurality of color filter areas corresponding to the plurality of subpixel areas, respectively, and the color conversion layer may include a plurality of color conversion areas corresponding to the plurality of subpixel areas, respectively.

(226) A pixel-defining layer may be between the plurality of subpixel areas to define each of the subpixel areas.

(227) The color filter may further include a plurality of color filter areas and a light-blocking pattern between the plurality of color filter areas, and the color conversion layer may further include a plurality of color conversion areas and a light-blocking pattern between the plurality of color conversion areas.

(228) The plurality of color filter areas (or the plurality of color conversion areas) may include a first area emitting first color light, a second area emitting second color light, and/or a third area emitting third color light, and the first color light, the second color light, and/or the third color light may have different maximum emission wavelengths from one another. In an embodiment, the first color light may be red light, the second color light may be green light, and the third color light may be blue light. In an embodiment, the plurality of color filter areas (or the plurality of color conversion areas) may include a quantum dot. In more detail, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dot is the same as described elsewhere in the present specification. The first area, the second area and/or the third area may further include a scatterer.

(229) In an embodiment, the light-emitting device may emit a first light, the first area may absorb the first light to emit a first first-color light, the second area may absorb the first light to emit a second first-color light, and the third area may absorb the first light to emit a third first-color light. In this regard, the first first-color light, the second first-color light, and the third first-color light may have different maximum emission wavelengths from one another. In more detail, the first light may be blue light, the first first-color light may be red light, the second first-color light may be green light, and the third first-color light may be blue light.

(230) The electronic apparatus may further include a thin-film transistor in addition to the light-emitting device as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one selected from the source electrode and the drain electrode may be electrically coupled to any one selected from the first electrode and the second electrode of the light-emitting device.

(231) The thin-film transistor may further include a gate electrode, a gate insulating film, and/or the like.

(232) The active layer may include crystalline silicon, amorphous silicon, organic semiconductor, oxide semiconductor, and/or the like.

(233) The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion may be between the color filter and/or the color conversion layer and the light-emitting device. The sealing portion allows light from the light-emitting device to be extracted to the outside, while concurrently (e.g., simultaneously) preventing or reducing penetration of ambient air and/or moisture into the light-emitting device. The sealing portion may be a sealing substrate including a transparent glass substrate and/or a plastic substrate. The sealing portion may be a thin film encapsulation layer including at least one layer of an organic layer and an inorganic layer. When the sealing portion is a thin film encapsulation layer, the electronic apparatus may be flexible.

(234) On the sealing portion, in addition to the color filter and/or the color conversion layer, various suitable functional layers may be further included according to the use of the electronic apparatus. The functional layers may include a touch screen layer, a polarizing layer, and/or the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, and/or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus for authenticating an individual by using biometric information of a biometric body (for example, a finger tip, a pupil, and/or the like).

(235) The authentication apparatus may further include, in addition to the light-emitting device, a biometric information collector.

(236) The electronic apparatus may be applied to various suitable displays, light sources, lighting, personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, and/or endoscope displays), fish finders, various suitable measuring instruments, meters (for example, meters for a vehicle, an aircraft, and/or a vessel), projectors, and/or the like.

(237) Description of FIGS. 2 and 3

(238) FIG. 2 is a cross-sectional view of a light-emitting apparatus according to an embodiment.

(239) The light-emitting apparatus of FIG. 2 includes a substrate **100**, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion **300** that seals the light-emitting device.

(240) The substrate **100** may be a flexible substrate, a glass substrate, and/or a metal substrate. A buffer layer **210** may be on the substrate **100**. The buffer layer **210** prevents or reduces the penetration of impurities through the substrate **100** and may provide a flat surface on the substrate **100**.

(241) A TFT may be on the buffer layer **210**. The TFT may include an active layer **220**, a gate electrode **240**, a source electrode **260**, and a drain electrode **270**.

(242) The active layer **220** may include an inorganic semiconductor such as silicon and/or polysilicon, an organic semiconductor, and/or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

(243) A gate insulating film **230** for insulating the active layer **220** from the gate electrode **240** may be on the active layer **220**, and the gate electrode **240** may be on the gate insulating film **230**.

(244) An interlayer insulating film **250** may be on the gate electrode **240**. The interlayer insulating film **250** is between the gate electrode **240** and the source electrode **260** to insulate the gate electrode **240** from the source electrode **260** and between the gate electrode **240** and the drain electrode **270** to insulate the gate electrode **240** from the drain electrode **270**.

(245) The source electrode **260** and the drain electrode **270** may be on the interlayer insulating film **250**. The interlayer insulating film **250** and the gate insulating film **230** may be formed to expose the source region and the drain region of the active layer **220**, and the source electrode **260** and the drain electrode **270** may be in contact with the exposed portions of the source region and the drain

region of the active layer **220**.

(246) The TFT may be electrically coupled to a light-emitting device to drive the light-emitting device, and is covered by a passivation layer **280**. The passivation layer **280** may include an inorganic insulating film, an organic insulating film, or a combination thereof. A light-emitting device is provided on the passivation layer **280**. The light-emitting device includes the first electrode **110**, the interlayer **130**, and the second electrode **150**.

(247) The first electrode **110** may be on the passivation layer **280**. The passivation layer **280** may not completely cover the drain electrode **270** and expose a portion of the drain electrode **270**, and the first electrode **110** may be coupled to the exposed portion of the drain electrode **270**.

(248) A pixel-defining layer **290** including an insulating material may be on the first electrode **110**. The pixel-defining layer **290** may expose a portion of the first electrode **110**, and the interlayer **130** may be formed in the exposed portion of the first electrode **110**. The pixel-defining layer **290** may be a polyimide and/or polyacrylic organic film. In one or more embodiments, at least some layers of the interlayer **130** may extend beyond the upper portion of the pixel-defining layer **290** and may thus be in the form of a common layer.

(249) The second electrode **150** may be on the interlayer **130**, and a capping layer **170** may be additionally formed on the second electrode **150**. The capping layer **170** may cover the second electrode **150**.

(250) The encapsulation portion **300** may be on the capping layer **170**. The encapsulation portion **300** may be on a light-emitting device and protects the light-emitting device from moisture and/or oxygen. The encapsulation portion **300** may include: an inorganic film including silicon nitride (SiN.sub.x), silicon oxide (SiO.sub.x), indium tin oxide, indium zinc oxide, or a combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (for example, polymethyl methacrylate and/or polyacrylic acid), an epoxy-based resin (for example, aliphatic glycidyl ether (AGE)), or a combination thereof; or a combination of an inorganic film and an organic film.

(251) FIG. **3** is a cross-sectional view showing a light-emitting apparatus according to an embodiment of the present disclosure.

(252) The light-emitting apparatus of FIG. **3** is the same as the light-emitting apparatus of FIG. **2**, except that a light-blocking pattern **500** and a functional region **400** are additionally on the encapsulation portion **300**. The functional region **400** may be i) a color filter area, ii) a color conversion area, or iii) a combination of the color filter area and the color conversion area. In an embodiment, the light-emitting device included in the light-emitting apparatus of FIG. **3** may be a tandem light-emitting device.

(253) Preparation Method

(254) Layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

(255) When layers constituting the hole transport region, an emission layer, and layers constituting the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10.sup.-8 torr to about 10.sup.-3 torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec by taking into account a material to be included in a layer to be formed and the structure of a layer to be formed.

Definition of at Least Some of the Terms

(256) The term “C.sub.3-C.sub.60 carbocyclic group,” as used herein, refers to a cyclic group that includes (e.g. consists of) only carbon as a ring-forming atom and has three to sixty carbon atoms, and the term “C.sub.1-C.sub.60 heterocyclic group,” as used herein, refers to a cyclic group that

has one to sixty carbon atoms and further has, in addition to carbon, a heteroatom as a ring-forming atom. The C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group may each be a monocyclic group that includes (e.g. consists of) one ring or a polycyclic group in which two or more rings are condensed with (e.g., combined together with) each other. In an embodiment, the number of ring-forming atoms of the C.sub.1-C.sub.60 heterocyclic group may be from 3 to 61. (257) The term “cyclic group,” as used herein, includes the C.sub.3-C.sub.60 carbocyclic group and the C.sub.1-C.sub.60 heterocyclic group.

(258) The term “ π electron-rich C.sub.3-C.sub.60 cyclic group,” as used herein, refers to a cyclic group that has three to sixty carbon atoms and does not include —N= as a ring-forming moiety, and the term “ π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group,” as used herein, refers to a heterocyclic group that has one to sixty carbon atoms and includes —N= as a ring-forming moiety.

(259) In an embodiment, the C.sub.3-C.sub.60 carbocyclic group may be i) a group T1 or ii) a condensed cyclic group in which two or more groups T1 are condensed with (e.g., combined together with) each other (for example, a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, acenaphthenyl group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group), the C.sub.1-C.sub.60 heterocyclic group may be i) a group T2, ii) a condensed cyclic group in which two or more groups T2 are condensed with (e.g., combined together with) each other, or iii) a condensed cyclic group in which at least one groups T2 and at least one group T1 are condensed with (e.g., combined together with) each other (for example, a pyrrole group, a thiophene group, a furan group, an indole group, a benzoindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, or an azadibenzofuran group), the π electron-rich C.sub.3-C.sub.60 cyclic group may be i) a group T1, ii) a condensed cyclic group in which two or more groups T1 are condensed with (e.g., combined together with) each other, iii) a group T3, iv) a condensed cyclic group in which two or more groups T3 are condensed with (e.g., combined together with) each other, or v) a condensed cyclic group in which at least one group T3 and at least one group T1 are condensed with (e.g., combined together with) each other (for example, the C.sub.3-C.sub.60 carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a

furan group, an indole group, a benzoindole group, a naphthoindole group, an isindole group, a benzoisindole group, a naphthoisindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoindolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.), the π -electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group may be i) a group T4, ii) a condensed cyclic group in which two or more groups T4 are condensed with (e.g., combined together with) each other, iii) a condensed cyclic group in which at least one group T4 and at least one group T1 are condensed with (e.g., combined together with) each other, iv) a condensed cyclic group in which at least one group T4 and at least one group T3 are condensed with (e.g., combined together with) each other, or v) a condensed cyclic group in which at least one group T4, at least one group T1, and at least one group T3 are condensed with (e.g., combined together with) each other (for example, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, or an azadibenzofuran group), the group T1 may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane group (or, a bicyclo[2.2.1]heptane group), a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, or a benzene group, the group T2 may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydro pyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group, the group T3 may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and the group T4 may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.

(260) The terms “the cyclic group,” “the C.sub.3-C.sub.60 carbocyclic group,” “the C.sub.1-C.sub.60 heterocyclic group,” “the π electron-rich C.sub.3-C.sub.60 cyclic group,” or “the π electron-deficient nitrogen-containing C.sub.1-C.sub.60 cyclic group,” as used herein, refer to a monovalent or pollyvalent group (for example, a divalent group, a trivalent group, a tetravalent group, or the like) that is condensed with (e.g., combined together with) a cyclic group, according

to the structure of a formula described with corresponding terms. In an embodiment, “a benzene group” may be a benzo group, a phenyl group, a phenylene group, or the like, which may be easily understood by one of ordinary skill in the art according to the structure of a formula including the “benzene group.”

(261) In an embodiment, examples of the monovalent C.sub.3-C.sub.60 carbocyclic group and the monovalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkyl group, a C.sub.1-C.sub.10 heterocycloalkyl group, a C.sub.3-C.sub.10 cycloalkenyl group, a C.sub.1-C.sub.10 heterocycloalkenyl group, a C.sub.6-C.sub.60 aryl group, a C.sub.1-C.sub.60 heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, and examples of the divalent C.sub.3-C.sub.60 carbocyclic group and the divalent C.sub.1-C.sub.60 heterocyclic group may include a C.sub.3-C.sub.10 cycloalkylene group, a C.sub.1-C.sub.10 heterocycloalkylene group, a C.sub.3-C.sub.10 cycloalkenylene group, a C.sub.1-C.sub.10 heterocycloalkenylene group, a C.sub.6-C.sub.60 arylene group, a C.sub.1-C.sub.60 heteroarylene group, a divalent non-aromatic condensed polycyclic group, and divalent non-aromatic condensed heteropolycyclic group.

(262) The term “C.sub.1-C.sub.60 alkyl group,” as used herein, refers to a linear or branched aliphatic hydrocarbon monovalent group having 1 to 60 carbon atoms, and examples thereof include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, and a tert-decyl group. The term “C.sub.1-C.sub.60 alkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.60 alkyl group.

(263) The term “C.sub.2-C.sub.60 alkenyl group,” as used herein, refers to a monovalent hydrocarbon group having at least one carbon-carbon double bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of a C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethenyl group, a propenyl group, and a butenyl group. The term “C.sub.2-C.sub.60 alkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.2-C.sub.60 alkenyl group.

(264) The term “C.sub.2-C.sub.60 alkynyl group,” as used herein, refers to a monovalent hydrocarbon group having at least one carbon-carbon triple bond at a main chain (e.g., in the middle) or at a terminal end (e.g., the terminus) of a C.sub.2-C.sub.60 alkyl group, and examples thereof include an ethynyl group and a propynyl group. The term “C.sub.2-C.sub.60 alkynylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.2-C.sub.60 alkynyl group.

(265) The term “C.sub.1-C.sub.60 alkoxy group,” as used herein, refers to a monovalent group represented by —OA.sub.101 (wherein A.sub.101 is the C.sub.1-C.sub.60 alkyl group), and examples thereof include a methoxy group, an ethoxy group, and an isopropoxy group.

(266) The term “C.sub.3-C.sub.10 cycloalkyl group,” as used herein, refers to a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or a bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, and a bicyclo[2.2.2]octyl group. The term “C.sub.3-C.sub.10 cycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.3-C.sub.10 cycloalkyl group.

(267) The term “C.sub.1-C.sub.10 heterocycloalkyl group,” as used herein, refers to a monovalent cyclic group that further includes, in addition to a carbon atom, at least one heteroatom as a ring-

forming atom and has 1 to 10 carbon atoms, and examples thereof include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothienyl group. The term “C.sub.1-C.sub.10 heterocycloalkylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.10 heterocycloalkyl group.

(268) The term “C.sub.3-C.sub.10 cycloalkenyl group,” as used herein, refers to a monovalent cyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity (e.g., is not aromatic), and examples thereof include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term “C.sub.3-C.sub.10 cycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.3-C.sub.10 cycloalkenyl group.

(269) The term “C.sub.1-C.sub.10 heterocycloalkenyl group,” as used herein, refers to a monovalent cyclic group that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, 1 to 10 carbon atoms, and at least one double bond in the cyclic structure thereof. Examples of the C.sub.1-C.sub.10 heterocycloalkenyl group include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothienyl group. The term “C.sub.1-C.sub.10 heterocycloalkenylene group,” as used herein, refers to a divalent group having substantially the same structure as the C.sub.1-C.sub.10 heterocycloalkenyl group.

(270) The term “C.sub.6-C.sub.60 aryl group,” as used herein, refers to a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term “C.sub.6-C.sub.60 arylene group,” as used herein, refers to a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Examples of the C.sub.6-C.sub.60 aryl group include a fluorenyl group, a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, and an ovalenyl group. When the C.sub.6-C.sub.60 aryl group and the C.sub.6-C.sub.60 arylene group each include two or more rings, the two or more rings may be fused to each other (e.g., combined together with each other).

(271) The term “C.sub.1-C.sub.60 heteroaryl group,” as used herein, refers to a monovalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. The term “C.sub.1-C.sub.60 heteroarylene group,” as used herein, refers to a divalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. Examples of the C.sub.1-C.sub.60 heteroaryl group include a carbazolyl group, a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, and a naphthyridinyl group. When the C.sub.1-C.sub.60 heteroaryl group and the C.sub.1-C.sub.60 heteroarylene group each include two or more rings, the two or more rings may be condensed with (e.g., combined together with) each other.

(272) The term “monovalent non-aromatic condensed polycyclic group,” as used herein, refers to a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed with (e.g., combined together with) each other, only carbon atoms as ring-forming atoms, and non-aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed polycyclic group include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, and an indenoanthracenyl group. The term “divalent non-aromatic condensed polycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed polycyclic group.

(273) The term “monovalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other (e.g., combined together with each other), at least one heteroatom other than carbon atoms, as a ring-forming atom, and non-aromaticity in its entire molecular structure (e.g., is not aromatic when considered as a whole). Examples of the monovalent non-aromatic condensed heteropolycyclic group include a pyrrolyl group, a thienyl group, a furanyl group, an indolyl group, a benzoindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group, a benzothienyl group, a benzofuranyl group, a carbazolyl group, a dibenzosilolyl group, a dibenzothienyl group, a dibenzofuranyl group, an azacarbazolyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothienyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indenocarbazolyl group, an indolocarbazolyl group, a benzofurocarbazolyl group, a benzothienocarbazolyl group, a benzosilolocarbazolyl group, a benzoindolocarbazolyl group, a benzocarbazolyl group, a benzonaphthofuranyl group, a benzonaphthothienyl group, a benzonaphthosilolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothienyl group, and a benzothienodibenzothienyl group. The term “divalent non-aromatic condensed heteropolycyclic group,” as used herein, refers to a divalent group having substantially the same structure as the monovalent non-aromatic condensed heteropolycyclic group.

(274) The term “C.sub.6-C.sub.60 aryloxy group,” as used herein, indicates —OA.sub.102 (wherein A.sub.102 is the C.sub.6-C.sub.60 aryl group), and the term “C.sub.6-C.sub.60 arylthio group,” as used herein, indicates —SA.sub.103 (wherein A.sub.103 is the C.sub.6-C.sub.60 aryl group).

(275) The term “C.sub.7-C.sub.60 arylalkyl group,” as used herein, indicates —A.sub.104A.sub.105 (wherein A.sub.104 is a C.sub.1-C.sub.54 alkylene group, and A.sub.105 is a C.sub.6-C.sub.59 aryl group), and the term “C.sub.2-C.sub.60 heteroarylalkyl group,” as used herein, indicates —A.sub.106A.sub.107 (wherein A.sub.106 is a C.sub.1-C.sub.59 alkylene group, and A.sub.107 is a C.sub.1-C.sub.59 heteroaryl group). The term “R.sub.10a,” as used herein, refers to: deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, or a C.sub.2-C.sub.60 heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)

(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32). Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23 and Q.sub.31 to Q.sub.33, as described herein, may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; C.sub.1-C.sub.60 alkyl group; C.sub.2-C.sub.60 alkenyl group; C.sub.2-C.sub.60 alkynyl group; C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.7-C.sub.60 arylalkyl group, or a C.sub.2-C.sub.60 heteroarylalkyl group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

(276) The term “heteroatom,” as used herein, refers to any atom other than a carbon atom. Examples of the heteroatom include O, S, N, P, Si, B, Ge, Se, or any combination thereof.

(277) A third-row transition metal in the present specification includes hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), and/or gold (Au).

(278) The term “Ph,” as used herein, refers to a phenyl group, the term “Me,” as used herein, refers to a methyl group, the term “Et,” as used herein, refers to an ethyl group, the term “tert-Bu” or “Bu.sup.t,” as used herein, refers to a tert-butyl group, and the term “OMe,” as used herein, refers to a methoxy group.

(279) The term “biphenyl group,” as used herein, refers to “a phenyl group substituted with a phenyl group.” In other words, the “biphenyl group” is a substituted phenyl group having a C.sub.6-C.sub.60 aryl group as a substituent.

(280) The term “terphenyl group,” as used herein, refers to “a phenyl group substituted with a biphenyl group.” In other words, the “terphenyl group” is a substituted phenyl group having, as a substituent, a C.sub.6-C.sub.60 aryl group substituted with a C.sub.6-C.sub.60 aryl group.

(281) * and *, as used herein, unless defined otherwise, each refer to a binding site to a neighboring atom in a corresponding formula or a moiety.

(282) Hereinafter, a compound according to embodiments and a light-emitting device according to embodiments will be described in more detail with reference to Synthesis Examples and Examples. The wording “B was used instead of A” used in describing Synthesis Examples refers to that an identical molar equivalent of B was used in place of A.

EXAMPLES

Synthesis Example 1: Synthesis of Compound 1

(283) ##STR00116##

1) Synthesis of Intermediate 1-1

(284) In order to consecutively proceed the following reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

(285) [Reactor 1-1] While 20 g of 1,3-dibromobenzene was stirred in 40 ml of a tetrahydrofuran (THF) solvent at -78°C ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was further stirred at -78°C . for an hour to thereby obtain Reaction solution 1-1.

(286) [Reactor 1-2] 21.5 g of dichlorodiphenylsilane was dissolved in 80 ml of THF and then stirred at -78°C . for 30 minutes, to thereby obtain Reaction solution 1-2. Reaction solution 1-1 was added dropwise to Reaction solution 1-2 at -78°C . using a cannula. Reaction solution 1-1 and Reaction solution 1-2 were stirred at -78°C . until the following Reaction solution 1-3 was prepared.

(287) [Reactor 1-3] While 20.9 g of 4-bromodibenzo[b,d]furan was dissolved in 40 ml of THF and then stirred at -78°C ., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at -78°C . for an hour to thereby obtain Reaction solution 1-3. Reaction solution 1-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at -78°C . using a cannula. Afterwards, a mixture of Reaction solution 1-1 to Reaction solution 1-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An

organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 30 g (yield of 70%) of Intermediate 1-1. Intermediate 1-1 was confirmed by LC-MS. (C.sub.30H.sub.21BrOSi: M+1 505.49)

(288) 2) Synthesis of Compound 1

(289) 2 g of Intermediate 1-1, 1.06 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 4 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent, and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.2 g (yield of 85%) of high-purity Compound 1. Compound 1 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 2: Synthesis of Compound 3

(290) ##STR00117##

1) Synthesis of Intermediate 3-1

(291) In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

(292) [Reactor 2-3] While 20.9 g of 2-bromodibenzo[b,d]furan was dissolved in 40 ml of THF and then stirred at -78° C., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at -78° C. for an hour to thereby obtain Reaction solution 2-3. Reaction solution 2-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at -78° C. using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 2-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 30 g (yield of 70%) of Intermediate 3-1. Intermediate 3-1 was confirmed by LC-MS.

(C.sub.30H.sub.21BrOSi: M+1 505.49)

(293) 2) Synthesis of Compound 3

(294) 2 g of Intermediate 3-1, 1.06 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 4 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent, and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.4 g (yield of 92%) of high-purity Compound 3. Compound 3 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 3: Synthesis of Compound 5

(295) ##STR00118##

1) Synthesis of Compound 5

(296) 2 g of Intermediate 1-1, 1.41 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 4 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent and then stirred at 90° C. for 12 hours, to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.5 g (yield of 85%) of high-purity Compound 5. Compound 5 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 4: Synthesis of Compound 7

(297) ##STR00119##

1) Synthesis of Compound 7

(298) 2 g of Intermediate 3-1, 1.41 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 4 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent and then stirred at 90° C. for 12

hours, to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.4 g (yield of 81%) of high-purity Compound 7. Compound 7 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 5: Synthesis of Compound 49

(299) ##STR00120##

1) Synthesis of Intermediate 49-1

(300) In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

(301) [Reactor 49-3] While 22.3 g of 4-bromodibenzo[b,d]thiophene was dissolved in 100 ml of THF and then stirred at -60° C., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at -60° C. for an hour to thereby obtain Reaction solution 49-3. Reaction solution 49-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at -60° C. using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 49-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 24 g (yield of 54%) of Intermediate 49-1. Intermediate 49-1 was confirmed by LC-MS. (C.sub.30H.sub.21BrSSi: M+1 521.55)

(302) 2) Synthesis of Compound 49

(303) 2 g of Intermediate 49-1, 1.02 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 3.8 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent, and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.2 g (yield of 85%) of high-purity Compound 49. Compound 49 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 6: Synthesis of Compound 51

(304) ##STR00121##

1) Synthesis of Intermediate 51-1

(305) In order to consecutively proceed the reaction, three reactors were prepared at the same time, and the reaction was carried out according to the following procedure.

(306) [Reactor 51-3] While 22.3 g of 2-bromodibenzo[b,d]thiophene was dissolved in 100 ml of THF and then stirred at -60° C., 33.9 ml of 2.5 M n-BuLi was slowly added dropwise thereto. After the dropwise addition, the mixture was stirred at -60° C. for an hour to thereby obtain Reaction solution 51-3. Reaction solution 51-3 was added dropwise to a mixture of Reaction solution 1-1 and Reaction solution 1-2 at -60° C. using a cannula. Afterwards, a mixture of Reaction solution 1-1, Reaction solution 1-2, and Reaction solution 51-3 was stirred at room temperature for 12 hours to thereby obtain a reactant. An organic layer was extracted from the reactant and dried. The mixed organic material was separated and purified by column chromatography to thereby obtain 25 g (yield of 57%) of Intermediate 51-1. Intermediate 51-1 was confirmed by LC-MS. (C.sub.30H.sub.21BrSSi: M+1 521.55)

(307) 2) Synthesis of Compound 51

(308) 2 g of Intermediate 51-1, 1.02 g of 2-chloro-4,6-diphenyl-1,3,5-triazine (CAS=3842-55-5), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 3.8 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent, and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.1 g (yield of 81%) of high-purity Compound 51. Compound 51 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 7: Synthesis of Compound 53

(309) ##STR00122##

1) Synthesis of Compound 53

(310) 2 g of Intermediate 49-1, 1.37 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 3.8 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.5 g (yield of 85%) of high-purity Compound 53. Compound 53 was confirmed by LC-MS and .sup.1H-NMR.

Synthesis Example 8: Synthesis of Compound 55

(311) ##STR00123##

1) Synthesis of Compound 55

(312) 2 g of Intermediate 51-1, 1.37 g of 9-(4-chloro-6-phenyl-1,3,5-triazin-2-yl)-9H-carbazole (CAS=1268244-56-9), and 0.22 g of Pd(PPh.sub.3).sub.4 were dissolved in 3.8 ml of a 2 M K.sub.2CO.sub.3 aqueous solution and 20 ml of a THF solvent and then stirred at 90° C. for 12 hours. After the reaction was completed, an organic layer obtained by extracting a reaction solution was dried. The mixed organic material was separated and purified by column chromatography, recrystallized, and then purified by sublimation, to thereby obtain 2.7 g (yield of 92%) of high-purity Compound 55. Compound 55 was confirmed by LC-MS and .sup.1H-NMR.

(313) .sup.1H NMR and MS/FAB results for the compounds synthesized according to Synthesis Examples 1 to 8 are shown in Table 1 below. Even compounds other than the compounds shown in Table 1 may be easily recognized by those skilled in the art by referring to the above synthesis routes and source materials.

(314) TABLE-US-00001 TABLE 1 MS/FAB Compound .sup.1H NMR (δ) Calc found 1 8.36(m, 5H), 8.08(d, 1H), 7.98(d, 657.22 657.85 1H), 7.88(s, 1H), 7.31-7.50(m, 23H) 3 8.36(m, 5H), 7.98(d, 1H), 7.88(s, 657.22 657.85 1H), 7.46(s, 2H), 7.31-7.70(m, 22H) 5 8.55(d, 1H), 8.36-8.38(m, 3H), 746.25 746.95 8.19(d, 1H), 8.08(d, 1H), 7.88- 7.98(m, 3H), 7.31-7.50(m, 25H) 7 8.55(d, 1H), 8.36-8.38(m, 3H), 746.25 746.95 8.19(d, 1H), 7.88-7.98(m, 3H), 7.77(s, 1H), 7.16-7.64(m, 25H) 49 8.55(d, 1H), 8.36-8.45(d, 6H), 673.20 673.91 7.88-7.93(m, 2H), 7.38-7.64(m, 22H) 51 8.36-8.45(d, 6H), 8.06(d, 2H), 673.20 673.91 7.88-7.93(m, 2H), 7.38-7.64(m, 21H) 53 8.55(d, 2H), 8.36-8.45(m, 4H), 762.23 763.01 8.19(d, 1H), 7.88-7.94(m, 3H), 7.16-7.64(m, 24H) 55 8.55(d, 1H), 8.36-8.45(m, 4H), 762.23 763.01 8.19(d, 1H), 7.46-8.06(m, 2H), 7.88-7.94(m, 3H), 7.16-7.64(m, 23H)

Example 1

(315) As an anode, a 15 Ω /cm.sup.2 (1,200 Å) ITO glass substrate available from Corning Inc. was cut to a size of 50 mm×50 mm×0.7 mm, sonicated using isopropyl alcohol and pure water each for 5 minutes, and then cleaned by irradiation of ultraviolet rays and exposure of ozone thereto for 30 minutes. The ITO glass substrate was provided to a vacuum deposition apparatus.

(316) N,N'-di(1-naphthyl)-N,N'-diphenylbenzidine (NPD) was vacuum-deposited on the glass substrate to form a hole injection layer having a thickness of 300 Å.

(317) TCTA was vacuum-deposited on the hole injection layer to form a first hole transport layer having a thickness of 200 Å, and then CzSi was vacuum-deposited on the first hole transport layer to form a second hole transport layer having a thickness of 100 Å.

(318) Compound 1 (host) and Ir(pmp).sub.3 (dopant) were co-deposited at a weight ratio of 92:8 on the second hole transport layer to form an emission layer having a thickness of 200 Å.

(319) Then, TSPO1 was deposited on the emission layer to form an electron transport layer having a thickness of 200 Å, and then, TPBi as a buffer electron transport compound was deposited on the electron transport layer to form a buffer layer having a thickness of 300 Å.

(320) LiF, which is an alkali metal halide, was deposited on the buffer layer to form an electron

injection layer having a thickness of 10 Å, and Al was vacuum-deposited thereon to form an LiF/Al electrode having a thickness of 3,000 Å, thereby completing manufacture of a light-emitting device.

(321) ##STR00124## ##STR00125##

Examples 2 to 8 and Comparative Examples 1 to 3

(322) Light-emitting devices were manufactured in substantially the same manner as in Example 1, except that Compounds shown in Table 2 were each used instead of Compound 1 in forming an emission layer.

Evaluation Example 1

(323) To evaluate characteristics of the light-emitting devices manufactured according to Examples 1 to 8 and Comparative Examples 1 to 3, the driving voltage at a current density of 10 mA/cm², luminescence efficiency, and maximum external quantum efficiency (EQE) thereof were measured. The driving voltage of a light-emitting device were measured using a source meter (Keithley Instrument Inc., 2400 series), and the maximum external quantum efficiency was measured using the external quantum efficiency measurement apparatus C9920-2-12 of Hamamatsu Photonics Inc. In evaluating the maximum external quantum efficiency, the luminance/current density was measured using a luminance meter that was calibrated for wavelength sensitivity, and the maximum external quantum efficiency was converted by assuming an angular luminance distribution (Lambertian) which introduced a perfect reflecting diffuser. Table 2 below shows the evaluation results of the characteristics of the organic light-emitting devices.

(324) TABLE-US-00002 TABLE 2 Maximum Host in Driving Luminescence external emission voltage efficiency quantum Emission layer (V) (Cd/A) efficiency (%) color Example 1 Compound 1 4.2 22.5 22.1 Blue Example 2 Compound 3 4.1 22.4 21.6 Blue Example 3 Compound 5 4.2 21.1 20.8 Blue Example 4 Compound 7 4.3 22.7 21.7 Blue Example 5 Compound 49 4.1 22.4 21.8 Blue Example 6 Compound 51 4.0 22.3 21.3 Blue Example 7 Compound 53 4.2 22.0 21.8 Blue Example 8 Compound 55 4.1 22.1 22.0 Blue Comparative mCP 4.8 16.3 15.2 Blue Example 1 Comparative Compound 4.9 17.8 16.9 Blue Example 2 A Comparative Compound 4.8 17.5 17.1 Blue Example 3 B     0       

(325) From Table 2, it can be seen that the light-emitting devices of Examples 1 to 8 have reduced driving voltage, increased luminescence efficiency, and increased maximum quantum efficiency, compared to the light-emitting devices of Comparative Examples 1 and 3.

(326) The heterocyclic compound may be utilized in manufacturing a light-emitting device having high efficiency and long lifespan, and the light-emitting device may be utilized in manufacturing a high-quality electronic apparatus having excellent luminance efficiency and long lifespan.

(327) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the drawings, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims, and equivalents thereof.

Claims

1. A heterocyclic compound represented by Formula 1: ##STR00137## wherein, in Formula 1, EW.sub.11 is —CN, d1 is 0 or 1, wherein EW.sub.1 in Formula 1 is an imidazole group, a pyrazole group, a thiazole group, an isothiazole group, an oxazole group, an isoxazole group, a pyridine

group, a pyrazine group, a pyridazine group, a pyrimidine group, an indazole group, a purine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a phthalazine group, a naphthyridine group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a cinnoline group, a phenanthridine group, an acridine group, a phenanthroline group, a phenazine group, a benzimidazole group, a benzoisothiazole group, a benzoxazole group, a benzoisoxazole group, a triazole group, a tetrazole group, an oxadiazole group, a triazine group, a thiadiazole group, an imidazopyridine group, an imidazopyrimidine group, an azacarbazole group, an azadibenzofuran group, an azadibenzothiophene group, an azadibenzosilole group, an acridine group, or a pyridopyrazine group, or T.sub.1 is a group represented by Formula 2, and b1 is an integer from 1 to 5, wherein, in Formula 2, X.sub.1 is O or S, L.sub.1 is a single bond, a C.sub.4-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a1 is an integer from 0 to 5, and Ar.sub.1 and Ar.sub.2 are each independently a π electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a, and wherein, in Formulae 1 and 2, R.sub.1 and R.sub.2 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), or —C(=O)(Q.sub.1), c1 in Formula 1 is an integer from 0 to 10, c2 in Formula 2 is an integer from 0 to 7, and R.sub.10a is: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-Coo aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a

biphenyl group, or any combination thereof, * in Formula 2 indicates a binding site to EW.sub.1 in Formula 1, and wherein when d1 is 0, either (i) b1 is an integer from 2 to 5, or (ii) c1 in Formula 1 is an integer from 2 to 10 and at least one R.sub.1 is a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a and at least one other R.sub.1 is a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a.

2. The heterocyclic compound of claim 1, wherein the heterocyclic compound is represented by Formula 1(a) or 1(b): ##STR00138## wherein, in Formulae 1(a) and 1(b), X.sub.11 is N or C(R.sub.11), X.sub.12 is N or C(R.sub.12), X.sub.13 is N or C(R.sub.13), at least one selected from X.sub.11 to X.sub.13 is N or C(CN), R.sub.11 to R.sub.15 are the same as described in connection with R.sub.1, and T.sub.11 and T.sub.12 are the same as described in connection with T.sub.1.

3. A heterocyclic compound represented by Formula 1(a) or 1(b): ##STR00139## wherein, in Formulae 1(a) and 1(b), X.sub.11 is N or C(R.sub.11), X.sub.12 is N or C(R.sub.12), X.sub.13 is N or C(R.sub.13), T.sub.11 and T.sub.12 are each a group represented by Formula 2, and wherein, in Formula 2, X.sub.1 is O or S, L.sub.1 is a single bond, a C.sub.4-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, or a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a1 is an integer from 0 to 5, and Ar.sub.1 and Ar.sub.2 are each independently a Ir electron-rich C.sub.3-C.sub.60 cyclic group unsubstituted or substituted with at least one R.sub.10a, and wherein, in Formulae 1(a), 1(b), and 2, R.sub.2 and R.sub.11 to R.sub.15 are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkenyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.2-C.sub.60 alkynyl group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 alkoxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.3-C.sub.60 carbocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 aryloxy group unsubstituted or substituted with at least one R.sub.10a, a C.sub.6-C.sub.60 arylthio group unsubstituted or substituted with at least one R.sub.10a, —Si(Q.sub.1)(Q.sub.2)(Q.sub.3), —B(Q.sub.1)(Q.sub.2), —P(Q.sub.1)(Q.sub.2), or —C(=O)(Q.sub.1), wherein at least one of R.sub.11 to R.sub.15 is a C.sub.3-C.sub.60 carboxylic group unsubstituted or substituted with at least one R.sub.10a and at least one other of R.sub.11 to R.sub.15 is a C.sub.1-C.sub.60 heterocyclic group unsubstituted or substituted with at least one R.sub.10a, c2 in Formula 2 is an integer from 0 to 7, and R.sub.10a is: deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group; a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, or a C.sub.1-C.sub.60 alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.11)(Q.sub.12)(Q.sub.13), —N(Q.sub.11)(Q.sub.12), —B(Q.sub.11)(Q.sub.12), —C(=O)(Q.sub.11), —S(=O).sub.2(Q.sub.11), —P(=O)(Q.sub.11)(Q.sub.12), or any combination thereof; a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, or a C.sub.6-C.sub.60 arylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C.sub.1-C.sub.60 alkyl group, a C.sub.2-C.sub.60 alkenyl group, a C.sub.2-C.sub.60 alkynyl group, a C.sub.1-C.sub.60 alkoxy group, a C.sub.3-C.sub.60 carbocyclic group, a C.sub.1-C.sub.60 heterocyclic group, a C.sub.6-C.sub.60 aryloxy group, a C.sub.6-C.sub.60 arylthio group, a C.sub.7-C.sub.60 arylalkyl group, a C.sub.2-C.sub.60 heteroarylalkyl group, —Si(Q.sub.21)(Q.sub.22)(Q.sub.23), —N(Q.sub.21)(Q.sub.22), —B(Q.sub.21)(Q.sub.22), —C(=O)(Q.sub.21), —S(=O).sub.2(Q.sub.21), —P(=O)(Q.sub.21)(Q.sub.22), or any combination thereof; or —Si(Q.sub.31)(Q.sub.32)(Q.sub.33), —N(Q.sub.31)

(Q.sub.32), —B(Q.sub.31)(Q.sub.32), —C(=O)(Q.sub.31), —S(=O).sub.2(Q.sub.31), or —P(=O)(Q.sub.31)(Q.sub.32), wherein Q.sub.1 to Q.sub.3, Q.sub.11 to Q.sub.13, Q.sub.21 to Q.sub.23, and Q.sub.31 to Q.sub.33 are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C.sub.1-C.sub.60 alkyl group; a C.sub.2-C.sub.60 alkenyl group; a C.sub.2-C.sub.60 alkynyl group; a C.sub.1-C.sub.60 alkoxy group; or a C.sub.3-C.sub.60 carbocyclic group or a C.sub.1-C.sub.60 heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C.sub.1-C.sub.60 alkyl group, a C.sub.1-C.sub.60 alkoxy group, a phenyl group, a biphenyl group, or any combination thereof, * in Formula 2 indicates a binding site to T.sub.11 or T.sub.12 in Formula 1(a) or Formula 1(b), and wherein at least two of X.sub.11 to X.sub.13 are each N.

4. The heterocyclic compound of claim 2, wherein X.sub.11 is C(R.sub.11), X.sub.12 is C(R.sub.12), X.sub.13 is C(R.sub.13), and at least one selected from X.sub.11 to X.sub.13 is C(CN).

5. The heterocyclic compound of claim 1, wherein b1 in Formula 1 is 1 or 2.

6. The heterocyclic compound of claim 1, wherein Formula 2 is represented by one selected from Formulae 2-1 to 2-4: ##STR00140## wherein, in Formulae 2-1 to 2-4, X.sub.1, L.sub.1, a1, Ar.sub.1, Ar.sub.2, R.sub.2, and c2 are the same as described in claim 1, and * indicates a binding site to EW.sub.1 in Formula 1.

7. The heterocyclic compound of claim 1, wherein L.sub.1 is: a single bond; or a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, or a 1,2,3,4-tetrahydronaphthalene group, each unsubstituted or substituted with at least one R.sub.10a.

8. The heterocyclic compound of claim 1, wherein Ar.sub.1 and Ar.sub.2 are each independently a benzene group, a heptalene group, an indene group, a naphthalene group, an azulene group, an indacene group, an acenaphthenyl group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, a naphthacene group, a picene group, a perylene group, a pentacene group, a hexacene group, a pentaphene group, a rubicene group, a coronene group, an ovalene group, a pyrrole group, a furan group, a thiophene group, an isoindole group, an indole group, an indene group, a benzofuran group, a benzothiophene group, a benzosilole group, a naphtho pyrrole group, a naphthofuran group, a naphthothiophene group, a naphthosilole group, a benzocarbazole group, a dibenzocarbazole group, a dibenzofuran group, a dibenzothiophene group, a carbazole group, a dibenzosilole group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a triindolobenzene group, a pyrrolophenanthrene group, a furano phenanthrene group, a thienophenanthrene group, a benzonaphthofuran group, a benzonaphthothiophene group, a (indolo) phenanthrene group, a (benzofurano) phenanthrene group, or a (benzothieno) phenanthrene group, each unsubstituted or substituted with at least one R.sub.10a.

9. A heterocyclic compound, wherein the heterocyclic compound is selected from Compounds 5 to 28, 37 to 48, 53 to 50 to 76 and 81-96: ##STR00141## ##STR00142## ##STR00143## ##STR00144## ##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149## ##STR00150## ##STR00151## ##STR00152## ##STR00153## ##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158## ##STR00159## ##STR00160##

10. The heterocyclic compound of claim 1, wherein the heterocyclic compound emits blue light having a maximum emission wavelength of 400 nm or more and 465 nm or less, and a difference between a singlet (S.sub.1) energy level and a triplet (T.sub.1) energy level of the heterocyclic compound is 0.5 eV or less.

11. A light-emitting device comprising: a first electrode; a second electrode facing the first electrode; an interlayer between the first electrode and the second electrode and including an

emission layer; and the heterocyclic compound represented by Formula 1 of claim 1.

12. The light-emitting device of claim 11, wherein: the first electrode is an anode, the second electrode is a cathode, the interlayer further comprises a hole transport region between the first electrode and the emission layer and an electron transport region between the emission layer and the second electrode, the hole transport region comprises a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region comprises a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

13. The light-emitting device of claim 12, wherein at least one selected from the emission layer and the electron transport region comprises the heterocyclic compound.

14. The light-emitting device of claim 11, wherein the emission layer comprises a host, and the heterocyclic compound acts as the host.

15. The light-emitting device of claim 12, further comprising: a first capping layer outside the first electrode and including the heterocyclic compound represented by Formula 1; a second capping layer outside the second electrode and including the heterocyclic compound represented by Formula 1; or the first capping layer and the second capping layer.

16. An electronic apparatus comprising the light-emitting device of claim 12.

17. The electronic apparatus of claim 16, further comprising a thin-film transistor, wherein the thin-film transistor comprises a source electrode and a drain electrode, and the first electrode of the light-emitting device is electrically coupled to at least one selected from the source electrode and the drain electrode of the thin-film transistor.

18. The electronic apparatus of claim 16, further comprising a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof.
